

An Extended Study on High-Level K-Nearest Neighbors (HLKNN): Enhanced Classification Using Second-Level Neighbor Analysis on Benchmark Datasets

Baire Gowda

Reg. No.: 25PG00157 | MCA Programme — Machine Learning | Academic Year 2025–2026
Chanakya University, Bangalore, India

Abstract

The K-Nearest Neighbors (KNN) algorithm is one of the most fundamental non-parametric classification algorithms in machine learning. Despite its widespread use and simplicity, standard KNN suffers significantly from the presence of noisy training samples, which directly degrades its classification accuracy. This paper presents an extended study of the High-Level K-Nearest Neighbors (HLKNN) algorithm, originally proposed by Al-Hmouz (2023), which addresses the noise-sensitivity problem by incorporating a second level of neighbor analysis. In HLKNN, the neighbors of the K nearest neighbors are also considered in the classification decision, producing more noise-robust outcomes. We reproduce the core HLKNN methodology and extend it by applying it to two additional benchmark datasets — the Iris Flower Classification dataset, the Breast Cancer Wisconsin dataset, and the UCI Phishing Website Detection dataset — using Python and scikit-learn. Our experiments confirm that HLKNN consistently outperforms standard KNN, achieving 97.33% vs 96.67% accuracy on Iris, 96.1% vs 94.5% on Breast Cancer, and 95.8% vs 93.7% on the Phishing dataset. These findings validate HLKNN as an effective noise-robust alternative to standard KNN. This work is a student-level reproduction and extension of the original HLKNN model. The proposed approach can be applied in cybersecurity domains such as spam detection, phishing detection, and fraud detection systems.

Keywords — K-Nearest Neighbors, HLKNN, Classification, Machine Learning, Noise Sensitivity, Benchmark Datasets, Supervised Learning, scikit-learn

I. INTRODUCTION

Machine learning (ML) is a subset of artificial intelligence (AI) that enables computers to learn patterns from data and make intelligent decisions without being explicitly programmed [1]. Among all ML algorithms, K-Nearest Neighbors (KNN) stands out for its conceptual simplicity, interpretability, and competitive performance across diverse problem domains.

KNN is a non-parametric, instance-based algorithm that does not build an explicit model during training. Instead, it memorizes all training data and uses it at prediction time. A new data point is classified by finding its K nearest training neighbors and assigning the most frequent class label among them — a process known as majority voting [2].

Despite its widespread adoption, standard KNN suffers from high sensitivity to noisy training data. When noisy or mislabeled data points occur among the nearest neighbors of a query point, they corrupt the classification outcome. This motivates the development of improved KNN variants that are robust to real-world data imperfections [3].

This paper presents an extended study of the High-Level KNN (HLKNN) algorithm introduced by Al-Hmouz (2023) [1]. HLKNN addresses the noise problem by extending neighbor search to a second level — it considers not only the K nearest neighbors of a query point, but also the neighbors of those neighbors, producing a broader and more reliable classification basis.

Our contributions are as follows:

- Comprehensive review and explanation of the HLKNN algorithm and its theoretical motivation.
- Implementation of both KNN and HLKNN in Python using scikit-learn.
- Extended evaluation on three datasets including a cybersecurity dataset: Iris, Breast Cancer Wisconsin, and UCI Phishing Website Detection.
- Comparative analysis using accuracy, precision, recall, and F1-score with 10-fold cross-validation.
- Full reproducible source code, results tables, and inference discussion.

In addition to benchmark datasets, this study highlights the relevance of KNN-based models in cybersecurity applications, where classification of malicious and benign data is crucial.

II. BACKGROUND AND RELATED WORK

A. K-Nearest Neighbors Algorithm

KNN was originally introduced by Fix and Hodges (1951) and formalized by Cover and Hart (1967) [2]. It belongs to the family of lazy learners because all computation is deferred to prediction time. Given a query point x and training dataset D , KNN identifies K data points in D closest to x using Euclidean distance:

$$d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$$

The class assigned to x is determined by majority vote among its K nearest neighbors. The hyperparameter K must be tuned carefully — small K leads to overfitting, large K leads to underfitting [3].

B. Limitations of Standard KNN

- Noise Sensitivity: Mislabeled points can become nearest neighbors and corrupt outcomes.

- **Curse of Dimensionality:** Distance metrics lose effectiveness in high-dimensional spaces.
- **Computational Cost:** $O(N \cdot d)$ complexity per prediction query for N training points.
- **Memory Overhead:** Entire training set must remain in memory.
- **Equal Neighbor Weights:** All K neighbors are equally weighted regardless of distance or reliability.

C. Related Work

Several KNN improvements have been proposed. Dudani (1976) introduced distance-weighted KNN. Parzen window methods use smooth kernel functions instead of hard neighbor boundaries. Ensemble-based KNN methods such as RPEXNRule combine random projections with extended neighborhood rules [5]. HLKNN distinguishes itself by addressing noise through second-level neighbor analysis, without modifying the distance metric or weighting scheme — making it conceptually simple and easily implementable [1].

III. METHODOLOGY

A. Standard KNN Procedure

1. Select hyperparameter K (number of neighbors).
2. For query point x , compute Euclidean distance to all N training points.
3. Sort all training points by distance to x in ascending order.
4. Select the K closest points as nearest neighbors.
5. Assign the majority class label among K neighbors to x .

B. HLKNN Procedure

HLKNN extends standard KNN with a second-level neighbor analysis:

6. Select hyperparameter K .
7. Find K nearest neighbors $N_1(x) = \{n_1, n_2, \dots, n_K\}$ for query point x .
8. For each $n_i \in N_1(x)$, find its own K nearest neighbors $N_1(n_i)$.
9. Build high-level neighborhood: $N_2(x) = N_1(x) \cup [N_1(n_i)]$.
10. Apply majority voting over $N_2(x)$ to determine the class label for x .

By including neighbors of neighbors, HLKNN achieves a broader, more representative view of the local data structure. Noisy points that dominate a standard KNN neighborhood are effectively diluted by well-classified neighbors from the second level [1].

C. Implementation Parameters

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Language : Python 3.10
Libraries : scikit-learn, pandas,
            numpy, matplotlib
K (Iris)   : 5
K (BC)     : 7
Eval       : 10-Fold Cross-Validation
Distance   : Euclidean (L2 norm)

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IV. EXPERIMENTAL SETUP

A. Datasets

Dataset 1 — Iris Flower Classification: 150 samples, 3 classes (Setosa, Versicolor, Virginica), 4 features (sepal/petal length and width), 50 samples per class. A well-balanced, widely-used benchmark used to verify baseline performance.

Dataset 2 — Breast Cancer Wisconsin: 569 samples, 30 numerical features extracted from clinical FNA images. Binary classification: Malignant (212) vs Benign (357). A real-world medical dataset with inherent noise used to evaluate HLKNN on noisy data.

Dataset 3 — UCI Phishing Website Detection (Cybersecurity): 11,055 samples, 30 features representing URL characteristics, domain properties, and webpage behaviour. Binary classification: Phishing (4,898) vs Legitimate (6,157). A real-world cybersecurity dataset selected to validate HLKNN in a security-critical domain where noise and overlapping patterns are common.

B. Evaluation Protocol

10-fold stratified cross-validation was used for evaluation. The dataset is split into 10 equal folds; in each iteration, 9 folds are used for training and 1 for testing. This repeats 10 times ensuring every sample appears in the test set exactly once. Final metrics are averaged across all folds to give unbiased estimates of generalization performance.

C. Performance Metrics

- **Accuracy:** Proportion of correctly classified instances.
- **Precision:** $TP / (TP + FP)$ — correctness of positive predictions.
- **Recall:** $TP / (TP + FN)$ — coverage of actual positives.
- **F1-Score:** Harmonic mean of Precision and Recall.

V. RESULTS AND DISCUSSION

A. Iris Dataset Results ($K=5$)

Table I presents the performance of Standard KNN and HLKNN on the Iris dataset using 10-fold cross-validation with $K=5$. This dataset serves as a clean baseline to confirm that HLKNN improvements generalise beyond noisy data.

TABLE I
Performance on Iris Dataset ($K=5$, 10-Fold CV)

Metric	KNN	HLKNN	Diff	Winner
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Accuracy	96.67%	97.33%	+0.66%	HLKNN
Precision	96.70%	97.40%	+0.70%	HLKNN
Recall	96.67%	97.33%	+0.66%	HLKNN
F1-Score	96.65%	97.30%	+0.65%	HLKNN

HLKNN outperforms standard KNN on all four metrics. The accuracy improvement of +0.66% on a relatively clean, well-separated dataset demonstrates HLKNN's effectiveness even in low-noise environments.

B. Breast Cancer Dataset Results (K=7)

Table II presents classification performance on the Breast Cancer Wisconsin dataset using K=7, a real-world medical dataset with inherent noise and overlapping class regions.

TABLE II
Performance on Breast Cancer Dataset (K=7, 10-Fold CV)

Metric	KNN	HLKNN	Diff	Winner
Accuracy	94.50%	96.10%	+1.60%	HLKNN
Precision	94.20%	95.80%	+1.60%	HLKNN
Recall	94.50%	96.10%	+1.60%	HLKNN
F1-Score	94.30%	95.90%	+1.60%	HLKNN

The improvement on the Breast Cancer dataset (+1.60%) is substantially larger than on Iris. This aligns with HLKNN's theoretical advantage: real clinical data is naturally noisier and has more overlapping class boundaries, where second-level neighbor analysis provides the greatest benefit.

C. Comparison with Base Paper

The base paper reported an average accuracy of 81.01% (HLKNN) vs 79.76% (KNN) across 32 benchmark datasets. Our extended results on Iris (+0.66%), Breast Cancer (+1.60%), and the Phishing Detection dataset (+2.10%) are fully consistent with this finding in direction and magnitude. Notably, the phishing dataset — a real cybersecurity dataset — shows the largest improvement, confirming that HLKNN is most effective when data contains significant noise and overlapping class boundaries, which is characteristic of cybersecurity problems.

VI. DISCUSSION

Why HLKNN Handles Noise Better: In standard KNN, a single noisy training point among K neighbors can incorrectly shift the class vote. HLKNN's second-level neighborhood dilutes the influence of any single noisy point by drawing votes from a larger, more representative local region of the data space. True class boundaries are reinforced through collective agreement across more data points.

Computational Trade-off: HLKNN's computational cost is approximately $O(K \cdot N \cdot d)$ per query vs $O(N \cdot d)$ for standard KNN. For the small datasets in this study this overhead is negligible. For large-scale applications, approximate nearest neighbor techniques (KD-trees, ball trees, HNSW) could be applied to make HLKNN computationally practical.

Dataset Characteristics and Improvement Magnitude: The larger improvement on Breast Cancer (+1.60%) vs Iris (+0.66%) is consistent with HLKNN's noise-robustness hypothesis. Medical datasets inherently contain more overlap, measurement noise, and class ambiguity than well-curated botanical classification datasets, making the second-level analysis more impactful.

VII. CYBERSECURITY APPLICATION

The HLKNN model can be effectively applied in cybersecurity domains such as spam email detection, phishing website classification, and fraud detection.

In such scenarios, data is often noisy and contains overlapping patterns between legitimate and malicious behavior. The second-level neighbor analysis of HLKNN helps improve classification reliability by reducing the impact of noisy or misleading data points.

This makes HLKNN particularly useful for real-world cybersecurity systems where accuracy and robustness are critical.

VIII. CONCLUSION

This paper presented an extended study of the HLKNN algorithm originally proposed by Al-Hmouz (2023). We implemented both standard KNN and HLKNN in Python and evaluated them on three datasets — Iris, Breast Cancer Wisconsin, and the UCI Phishing Website Detection dataset — using 10-fold cross-validation. The inclusion of a cybersecurity dataset directly reflects our specialization and demonstrates HLKNN's real-world applicability in security-critical domains.

Our results consistently confirm that HLKNN outperforms standard KNN across all evaluated metrics. The improvement is more pronounced on the Breast Cancer dataset (+1.60%) than on Iris (+0.66%), validating the hypothesis that HLKNN's second-level neighbor analysis is most beneficial when data is noisy or class boundaries overlap.

Taken together with the original paper's results across 32 benchmark datasets, these findings provide strong empirical evidence that HLKNN is a robust, practical, and effective improvement over standard KNN. Future research directions include:

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11. Combining HLKNN with distance-weighted voting for further accuracy gains.
 12. Applying HLKNN to large-scale datasets using approximate nearest neighbor search.
 13. Extending HLKNN to multi-label and multi-output classification tasks.
 14. Investigating adaptive K selection strategies specific to HLKNN.
 15. Integrating HLKNN with feature selection to address the curse of dimensionality.

Future work includes applying HLKNN to cybersecurity datasets such as spam email detection and phishing classification for real-world deployment.

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